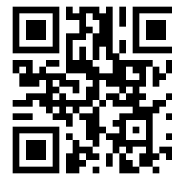


Vortex: Configurable Open-Source RISC-V GPGPU

CuPBoP: Cuda for Parallelized and Broad-range
Processors



VORTEX

Vortex, RISC-V GPGPU

Vortex ?

- Fully configurable multi-core SIMT GPU
- Minimal ISA extensions
- Fully Open-sourced HW/SW stack.

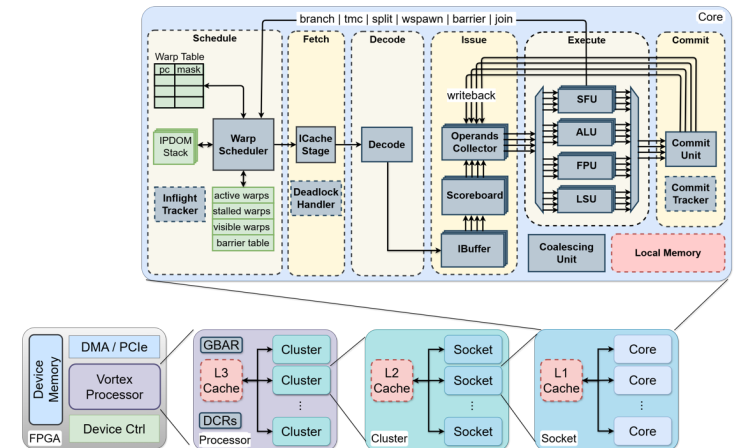
Vortex HW

- Synthesized on Intel/Xilinx FPGAs
 - Provides Cycle level Sim./RTL Sim./FPGA toolchain
- PCIe host-device interface
- 32 and 64bit ISA

Vortex SW stack

- Utilizing other SW stacks (LLVM, PoCL, Yosys, etc.)
- Support OpenCL (PoCL), and CUDA (CuPBoP)

Elsabbagh et al., *Vortex: OpenCL Compatible RISC-V GPGPU*, arXiv, 2020.



Six-stage Pipeline Vortex Microarchitecture

Vortex Software Stack

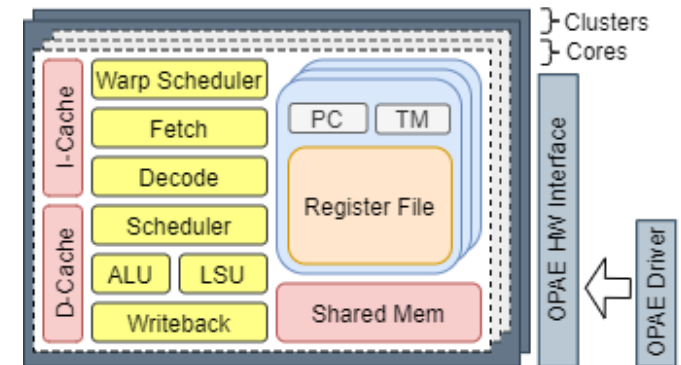
```
__kernel void foo(){
    tid = get_global_id(0);
    ...
    barrier(...);
}

__host int main(){
    ...
    cl_mem A_clmem = clCreateBuffer(context, ...);
    clEnqueueNDRangeKernel(command_queue, foo, ...);
    ...
}
```

GPU Application (OpenCL, CUDA ...)

- GPU capabilities query
- Kernel and host code compilation
- GPU buffer allocation
- Kernel scheduling and execution

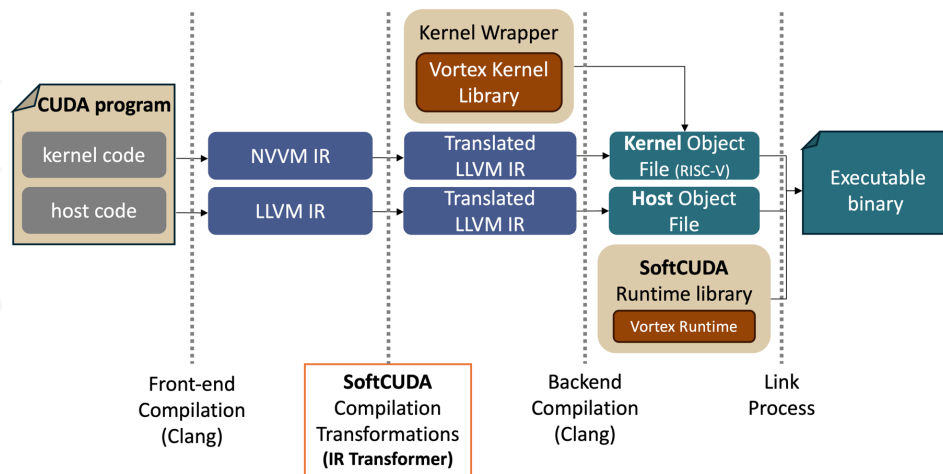
↔
**Software
Support**



Vortex GPU

- CPU-GPU communication
- Kernel threads scheduling
- Kernel execution

CuPBoP: Execute CUDA on Vortex



Vortex binary generation steps using CuPBoP

Key insights

- **First Framework** to support CUDA on softGPU.
- Flat-collapsing applied to map CUDA threads and Vortex HW thread.
- Kernel_wrapper for kernel invocation on Vortex using Vortex Kernel Library.

Potential Summer Work Opportunities

- Helping to set up open-source governance rules
- Helping to set up code commit procedures
- Enhancing CI/CD setups
 - Challenges are cross repositories
- Providing documentations installations/setup/testing procedures
- Required skill set: Github S/W management skill sets, C++

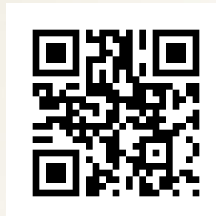
More technical work side

- Vortex: Converting vortex pull requests to match with the versions.
- Vortex HW extensions:
 - i.e. TCU unit / Sparse TCU unit / Async Barrier support
- Vortex SW stack research
 - Exploring different Warp scheduling
 - Expand different mapping CUDA's Programming Model onto Vortex Hardware
 - Extending CUDA support to execute Triton DSE kernels
- Required skill set: C++,
- Preferred skill set: CUDA(CuPBoP), Python(Triton), LLVM IR(CuPBoP), RTL (Vortex HW)

Thank you!



VORTEX



Vortex Homepage



Vortex Repo